# Supplementary Information

for: Landfill methane recovery and use under deep uncertainty: a satellite-constrained Monte Carlo decision model for the Norte III landfill, Buenos Aires

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## S1. Identifiability: variance shares, contraction and dependence

Eq. (12) of the manuscript gives the 2022 collection efficiency from the satellite emission, the oxidation fraction and the reported capture. It holds to 1e-16 in the sampled draws. Because the log-odds of capture is a sum of three terms, its variance divides into shares: in the prior (draws consistent with $\eta \le 0.95$) the emission carries 97% of the variance of $\ln[(1-\eta)/\eta]$ (standard deviation 0.41), the reported capture 3.4% and oxidation −0.0%; in the lognormal posterior the shares are 89%, 7% and 3.9% (standard deviation 0.27). Table S1 gives the prior-to-posterior contraction of the inputs that enter generation; contraction is 1 − IQR(posterior)/IQR(prior), and KS is the Kolmogorov–Smirnov distance between the two marginals.

| Input | Prior, median (P5–P95) | Posterior, median (P5–P95) | Contraction | KS |
|---|---|---|---|---|
| **Waste acceptance A (t/d)** | 16,500 (14,791–18,209) | 16,561 (14,862–18,243) | 1% | 0.026 |
| **Methane potential L0 (Nm³ $CH_4$/t)** | 105 (67–150) | 113 (79–151) | 13% | 0.138 |
| **Steady-state fraction f_ss** | 0.61 (0.40–0.84) | 0.65 (0.46–0.85) | 13% | 0.130 |
| **Methane fraction** | 0.50 (0.44–0.55) | 0.50 (0.44–0.55) | −0% | 0.010 |
| **Oxidation fraction** | 0.18 (0.09–0.30) | 0.17 (0.09–0.29) | 4% | 0.033 |
| **Collected biogas 2022 (Nm³/h)** | 13,049 (12,332–13,700) | 13,047 (12,329–13,702) | −1% | 0.006 |
| **Generation G22 (Nm³ $CH_4$/h)** | 43,527 (24,087–72,266) | 49,642 (34,300–70,876) | 26% | 0.238 |
| **Efficiency η 2022** | 0.15 (0.09–0.27) | 0.13 (0.09–0.19) | 43% | 0.231 |

*Table S1. Prior-to-posterior contraction (prior: draws consistent with $\eta \le 0.95$; posterior: lognormal).*

Posterior correlations (Pearson, of logarithms for the generation terms) are: ln L0 with ln f_ss −0.33 (prior 0.00), ln G22 with η22 −0.94, ln E22 with η22 −0.92, ln L0 with η22 −0.55 and ln f_ss with η22 −0.50. The negative correlation between L0 and f_ss is the ridge along which generation is constant; the satellite constraint selects the ridge, not a point on it.

## S2. Photovoltaic coating against conventional photovoltaics

On the covered area, conventional PV has a positive NPV in 4.0% of draws and a median LCOE of US$119 (84 to 169) per MWh, above the merchant value of electricity because of the landfill foundation premium, the capital premium and the discount rate. The paint, with deliberately wide priors (efficiency 2 to 10%, installed cost US$5 to 150 per m², life 2 to 20 years), has a median LCOE of US$211 (60 to 605) per MWh and is cheaper in 23% of draws. Setting the levelised cost of the paint equal to that of conventional PV gives the break-even efficiency

$$\eta^{*} = \frac{c_{pt}\ \kappa\ \left[CRF\left(r, L_{pt}\right) + o_{pt}\right]}{LCO\, E_{\mathrm{conv}}\ H_{poa}\ PR} \tag{S1}$$

which has a median of 9.7% (2.9% to 24.9%) and exceeds 10% in 48% of draws (Fig. S2a). The 23% figure is a property of unvalidated priors; the transferable result is the $\eta^{*}$ surface. Verified against the simulation: setting the paint efficiency to $\eta^{*}$ gives LCOE parity to numerical precision (automated test).

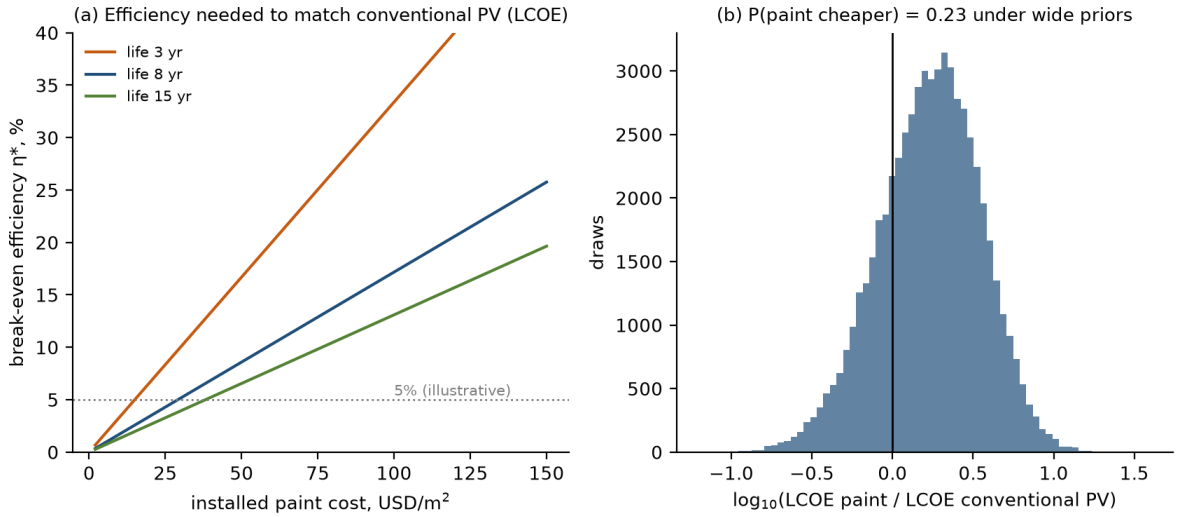



*Fig. S2. (a) Break-even efficiency of PV paint for LCOE parity with conventional PV against installed cost, for three lifetimes (other inputs at posterior medians). (b) Distribution of the LCOE ratio.*

## S3. Input distributions

Table S2 lists the sampled inputs. Distributions are triangular (minimum, mode, maximum), uniform (minimum, maximum) or lognormal (σ, mean 1). Tags: R, operator-reported or published programme value; L, literature or standard value; B, dated market or regulatory benchmark (Table 2 of the manuscript); A, assumed by the author and requiring a site-specific or dated quotation before use in a decision. Inputs marked * enter only the robustness variants. The sensitivities of Sections 3.8 and 3.9 of the manuscript (alternative generation priors, satellite scaling, engine slip 1 to 6%, marginal grid factor 0.35 to 0.65 with mode 0.45, counterfactual flare destruction 0.80 to 0.98, displacement 0.3 to 1.0) are tag A and are defined in NorteIII_review.py.

| Input | Description | Distribution | Tag |
|---|---|---|---|
| **Qcoll22** | collected biogas around 2022 (satellite period), Nm3/h; 13,100 peak in 2013, 13,800 in 2025 | tri(12,000, 13,100, 14,000) | A |
| **Awaste** | waste to Norte III, t/day (~18,000 reported) | tri(14,000, 16,500, 19,000) | R |
| **L0** | CH4 generation potential, Nm3 CH4 per t waste (inventory 100, CAA 170) | tri(50, 100, 170) | L |
| **fss** | fraction of steady-state generation reached (age/decay/lag) | tri(0.3, 0.6, 0.95) | A |
| **xCH4** | CH4 fraction of landfill gas | tri(0.42, 0.5, 0.57) | A |
| **ox** | cover methane oxidation fraction | tri(0.05, 0.15, 0.35) | L |
| **g_t** | generation 2025-26 relative to 2022 | tri(0.95, 1.05, 1.15) | A |
| **Qcoll** | collected biogas 2025-26, Nm3/h (13,800 in 2025; 16,000 Apr 2026) | tri(13,000, 14,800, 16,500) | R |
| **P_el** | installed biogas power, MW (23 in 2020; 33 planned 2026) | tri(23, 28, 33) | R |
| **CF_el** | engine capacity factor | tri(0.7, 0.85, 0.93) | A |
| **eta_el** | engine electrical efficiency | tri(0.36, 0.4, 0.43) | L |
| **d_f** | flare/engine CH4 destruction efficiency | tri(0.95, 0.98, 0.995) | L |
| **dEta1** | S1 absolute gain in collection efficiency (of generation) | tri(0, 0.02, 0.06) | A |
| **capex_s1** | S1 capex, MUSD | tri(0.5, 1.5, 4) | A |
| **opex_s1** | S1 opex, MUSD/yr | tri(0.15, 0.4, 1) | A |
| **f_hot** | flux fraction passing covered area | tri(0.1, 0.25, 0.5) | A |
| **h_e** | hot-spot flux enrichment over mean | tri(1.5, 3, 6) | A |
| **A_act** | actively emitting area, ha | tri(100, 200, 400) | A |
| **eps_conv** | conventional cover: fraction of arriving CH4 blocked | tri(0.3, 0.6, 0.85) | A |
| **eps_coat** | experimental coating: fraction blocked | tri(0.1, 0.45, 0.85) | A |
| **s_c** | share of blocked CH4 captured by extraction | tri(0.05, 0.35, 0.75) | A |
| **delta_ox** | fractional loss of oxidation under cover | tri(0.2, 0.6, 1) | A |

| Input | Description | Distribution | Tag |
|---|---|---|---|
| **rho_r** | oxidation of redirected gas relative to ox | uni(0, 1) | A |
| **c_conv** | conventional cover cost, kUSD/ha | tri(80, 150, 300) | A |
| **c_coat** | experimental coating cost, kUSD/ha | tri(100, 250, 700) | A |
| **om_cov** | cover O&M, fraction of capex per yr | tri(0.015, 0.03, 0.05) | A |
| **life_conv** | conventional cover life, yr | tri(15, 25, 30) | A |
| **life_coat** | coating life, yr | tri(4, 8, 15) | A |
| **r** | real USD discount rate (EMBI 2025: 561 to >1,200 bps; 569 bps Dec 2025) | tri(0.08, 0.12, 0.18) | B |
| **f_imp** | imported share of equipment capex | tri(0.5, 0.7, 0.9) | A |
| **duty** | import duty on capital goods (0% BK regime; 12.6% Decree 513/2025; 20% Mercosur CET) | tri(0, 0.126, 0.2) | B |
| **logist** | freight, logistics and FX cost on imports, fraction | tri(0.03, 0.08, 0.15) | A |
| **m_p** | realised-price multiplier (tariff/FX risk), mean 1 | logn(0.25) | A |
| **avail_firm** | fraction of surplus gas contractually firm | tri(0.6, 0.85, 1) | A |
| **uptime** | plant availability | tri(0.8, 0.9, 0.95) | A |
| **rec** | CH4 recovery in upgrading | tri(0.9, 0.95, 0.985) | L |
| **tail_ox** | fraction of tail-gas CH4 slip oxidised | uni(0, 0.95) | A |
| **fug_plant** | plant fugitive CH4, fraction of CH4 in | tri(0.002, 0.006, 0.012) | L |
| **capex_up** | upgrader capex, USD per Nm3/h raw capacity | tri(1,800, 3,000, 5,000) | A |
| **e_up** | upgrading electricity, kWh/Nm3 raw | tri(0.2, 0.28, 0.4) | L |
| **c_el** | electricity purchase cost, USD/MWh (wholesale monomic ~87-91 USD/MWh 2024-25 plus network) | tri(70, 95, 130) | B |
| **om_up** | upgrader O&M, fraction of capex per yr | tri(0.03, 0.04, 0.06) | A |
| **lab_up** | labour/other, MUSD/yr | tri(0.15, 0.25, 0.45) | A |
| **capex_gv** | GNC compression capex, USD per Nm3/h product | tri(400, 800, 1,500) | A |
| **capex_gf** | GNC fixed capex, MUSD | tri(0.3, 0.8, 2) | A |
| **p_gnc** | bio-GNC net plant-gate price, USD/Nm3 (delivered station gas ~0.15, retail ~0.40-0.45, 2025) | tri(0.13, 0.19, 0.4) | B |
| **p_pipe** | pipeline net plant-gate price, USD/Nm3 (PIST 2.94-2.98 USD/MMBtu, Dec 2025) | tri(0.07, 0.1, 0.16) | B |
| **D_gnc** | bio-GNC anchor demand, Nm3/h | tri(300, 1,000, 3,000) | A |
| **dist_km** | distance to injection point, km | tri(2, 8, 20) | A |
| **c_km** | pipeline spur cost, MUSD/km | tri(0.3, 0.6, 1.2) | A |
| **c_inj** | injection/metering station, MUSD | tri(0.5, 1, 2) | A |
| **capex_pw** | gas-engine power capex, USD/kW (RenovAr landfill-gas ref 1.3 MUSD/MW) | tri(1,100, 1,300, 1,700) | R |
| **p_el** | electricity sale price, merchant/MAT, USD/MWh (marginal cost 69, monomic ~87-91, 2024-25) | tri(60, 75, 95) | B |
| **p_el_k*** | electricity sale price, RenovAr-type contract, USD/MWh (landfill-gas awards 128-130, 2017) | tri(110, 125, 135) | B |
| **cf_pw** | new engines capacity factor | tri(0.8, 0.88, 0.93) | A |
| **eta_pw** | new engines electrical efficiency | tri(0.36, 0.4, 0.43) | L |
| **om_pw** | engine O&M, fraction of capex per yr | tri(0.03, 0.045, 0.07) | A |
| **var_loss** | flow-variability loss without buffer | uni(0.05, 0.25) | A |
| **zeta** | fraction of variability loss removed by buffer | uni(0.5, 0.9) | A |
| **capex_buf** | buffer capex, MUSD | tri(0.3, 0.8, 2) | A |
| **leak_buf** | buffer CH4 leakage, fraction of throughput | tri(0.002, 0.005, 0.01) | A |
| **f_pv** | share of upgrader electricity from on-site PV | uni(0.1, 0.5) | A |
| **dEta_ad** | adaptive-control gain in collection efficiency | tri(0, 0.01, 0.03) | A |
| **capex_ad** | adaptive control capex, MUSD | tri(0.2, 0.5, 1.2) | A |
| **opex_ad** | adaptive control opex, MUSD/yr | tri(0.05, 0.1, 0.25) | A |
| **kwp_ha** | conventional PV, kWp per ha | tri(500, 700, 900) | A |
| **capex_pv** | PV capex, USD/kW (RenovAr ref 850) | tri(550, 750, 1,100) | R |
| **prem_pv** | landfill foundation premium for PV | tri(1.1, 1.3, 1.6) | A |
| **yield_pv** | PV yield, kWh/kWp/yr | tri(1,350, 1,500, 1,650) | A |
| **om_pv** | PV O&M, fraction of capex per yr | tri(0.01, 0.015, 0.025) | A |
| **life_pv** | PV life, yr | tri(20, 25, 30) | A |
| **p_pv** | PV electricity value, merchant, USD/MWh (2024-25 wholesale cost) | tri(55, 70, 90) | B |
| **eta_pt** | PV-paint module efficiency (unvalidated) | tri(0.02, 0.05, 0.1) | A |
| **c_pt** | PV-paint installed cost, USD/m2 (unvalidated; wide on purpose) | tri(5, 30, 150) | A |
| **life_pt** | PV-paint life, yr (unvalidated; wide on purpose) | tri(2, 6, 20) | A |

| Input | Description | Distribution | Tag |
|---|---|---|---|
| **pr_pt** | PV-paint performance ratio | tri(0.65, 0.78, 0.85) | A |
| **om_pt** | PV-paint O&M, fraction of capex per yr | tri(0.01, 0.02, 0.04) | A |
| **H_poa** | plane-of-array irradiation, kWh/m2/yr | tri(1,550, 1,650, 1,750) | A |
| **ef_grid** | grid emission factor, tCO2/MWh | tri(0.28, 0.31, 0.36) | A |
| **cp*** | carbon-credit price if eligible, USD/tCO2e (VCM ~5.5, landfill gas ~9.5 USD, 2025) | tri(2, 6, 12) | B |
| **p_elig*** | probability that added capture is eligible for credits | uni(0.2, 0.8) | A |
| **dv*** | debt share of financing | tri(0, 0.4, 0.6) | A |
| **r_e*** | real cost of equity | tri(0.12, 0.16, 0.22) | A |
| **r_d*** | real cost of debt | tri(0.07, 0.09, 0.12) | A |
| **k_dec*** | first-order decay rate, 1/yr (IPCC 2006, temperate wet MSW) | tri(0.05, 0.09, 0.15) | L |
| **age*** | years of deposition before 2026 | tri(20, 28, 32) | A |
| **t_close*** | years until deposition stops, from 2026 | tri(3, 8, 15) | A |

*Table S2. Input distributions.*

## S4. Balances and automated tests

| Scenario | Emitted | Oxidised | Destroyed | Exported | Biogenic C | Exported energy | Electricity | Total $CO_2e$ |
|---|---|---|---|---|---|---|---|---|
| **S0** | 231.5 | 47.7 | 44.8 | 0.0 | 69.2 | 0.0 | 0.0 | 6.25 |
| **S1** | 224.6 | 46.2 | 53.2 | 0.0 | 74.5 | 0.0 | 0.0 | 6.06 |
| **S2** | 215.3 | 39.1 | 68.9 | 0.0 | 82.0 | 0.0 | 0.0 | 5.81 |
| **S4** | 215.3 | 39.1 | 68.9 | 0.0 | 82.0 | 0.0 | 0.0 | 5.81 |
| **S6** | 215.4 | 39.1 | 64.0 | 5.0 | 78.3 | 69.4 | 6.3 | 5.80 |
| **S6b** | 224.7 | 46.2 | 48.2 | 5.0 | 70.9 | 69.2 | 6.2 | 6.06 |
| **S7** | 211.9 | 38.3 | 68.3 | 5.0 | 80.9 | 69.4 | 4.3 | 5.71 |

*Table S3. Scenario balances for the median draw: methane emitted, oxidised, destroyed and exported (kt $CH_4$ $yr^{-1}$); biogenic carbon released as $CO_2$ (kt C $yr^{-1}$); biomethane energy exported and grid electricity used (GWh $yr^{-1}$); total emissions (Mt $CO_2e$ $yr^{-1}$, GWP100). Residuals of all balances are below $10^{-9}$ of generation.*

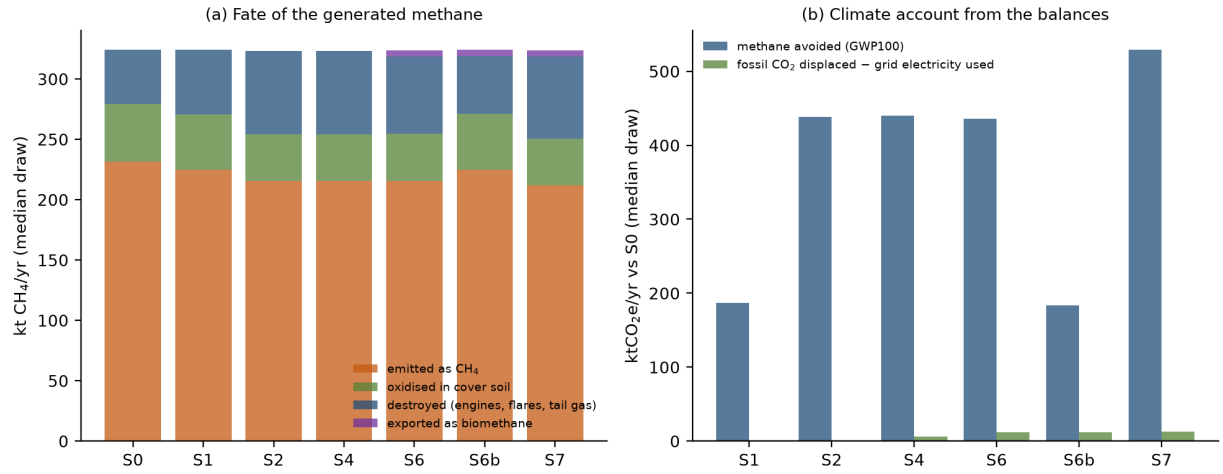



*Fig. S3. (a) Fate of the generated methane by scenario (median draw). (b) Climate account from the balances: methane avoided (GWP100) and fossil $CO_2$ displaced net of grid electricity used, relative to S0.*

The automated test suite (python NorteIII_MC.py --test) contains 59 checks, all of which pass: unit conversions (methane mass, carbon content, annuity and capital recovery factors); Latin-hypercube moments and the induced rank correlation; the white paper's screening arithmetic; closure of the cover balance; the analytic break-even $s_c^*$; two limiting cases of the cover; methane and carbon closure of seven scenario balances; methane and energy

closure of four upgrader accounts; equality of balance-based and incremental climate benefits for four scenarios; a hand-calculated upgrading case; an explicit cash-flow benchmark of the annuity NPV; unprofitability at zero gas prices; LCOE parity at $\eta^*$; the calibration window; physical consistency of the posterior; determinism; the steady-state limit of the time-resolved NPV and the analytic slow-decay limit of the first-order-decay model; the closed-form identity of the collection efficiency and the additive identity of its log-odds; the uninformative limit of the observation model; the reproduction of the annuity NPV by the engine counterfactual in the constant-supply and fast-decay limits; the capital-and-O&M floor under full curtailment; the zero of the engines' climate benefit at the break-even slip; the consistency of the conditional and unconditional probabilities; and the Monte Carlo standard error.

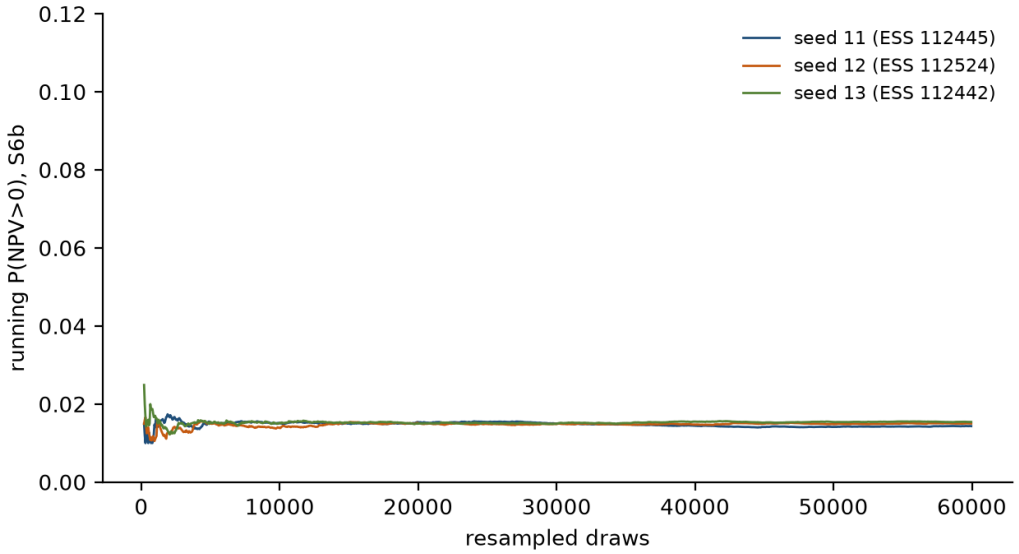



*Fig. S1. Running probability of positive NPV for S6b across three independent seeds (resampled lognormal posterior; ESS in the legend).*

## S5. Scorecard for the research hypotheses

| Hypothesis | Result |
|---|---|
| **H1: cover plus pressure-managed extraction lowers total emissions** | Supported conditionally: net reduction in 92% of draws if $s_c > s_c^*$ (median 0.14); net benefit is 64% of gross capture. |
| **H2: PV coatings beat conventional PV only above measurable thresholds** | Paint cheaper in 23% of draws under wide, unvalidated priors; break-even efficiency 9.7% at median inputs. |
| **H3: biomethane from surplus gas has higher risk-adjusted NPV than alternatives** | Not supported: beats additional engines in 0.02% of draws; break-even US$0.46 per Nm³ (US$14 per MMBtu). |
| **H4: buffering offsets its leakage, energy and capital costs** | Not supported: positive in 0.03% of draws; median ΔNPV −1.18 MUSD. |
| **H5: the integrated configuration has lower lifecycle $CO_2e$ per tonne of waste** | Supported but small: S7 lower in 99.6% of draws; median reduction 8.3%, from 1.04 to 0.95 $tCO_2e$ per tonne. |

*Table S4. Outcome of the five hypotheses of the project white paper (lognormal posterior).*